Lifecycle Marketing & CRM Analytics

Executive Summary

This project analyzes four lifecycle marketing campaigns and an always-on SMS reactivation program for a live entertainment and social platform.

Using SQL and Tableau, I examined how users moved through campaign funnels, how engagement and monetization differed across user segments, and where opportunities existed to improve CRM performance.

Key Findings

1. **Engagement:** Puzzle attracted 750 participants, but only 37.3% completed the event, highlighting a need for stronger progression incentives.
2. **Retention:** 94.3% of Check-in visitors unlocked the campaign, but only 37.5% completed all seven check-ins, revealing a sustained engagement challenge.
3. **Win-back:** King's Return reactivated 64.9% of eligible dormant users, while only 20.5% of gift claimers reached the 30-day login milestone.
4. **Monetization:** Fortune Recharge generated 164,294 in observed recharge revenue, with 59.3% contributed by just 29 users in the 1,500+ spend tier.
5. **SMS Reactivation:** Reactivation declined from 46.6% at three days of inactivity to 8.1% at 30 days, supporting earlier win-back intervention.

Business Recommendations

- Optimize campaign-specific funnel bottlenecks instead of applying one universal engagement strategy.
- Use historical user value to guide segmentation and personalized CRM incentives.
- Introduce progressive rewards and reminders to improve sustained engagement.
- Prioritize early SMS intervention and use targeted incentive-based campaigns for deeper or high-value win-back.
- Introduce control groups to measure incremental campaign impact.

Data note: The portfolio uses synthetically reconstructed data based on real campaign mechanics. Findings demonstrate the analytical methodology and should not be interpreted as experimentally verified causal effects.

Tableau Dashboard Gallery

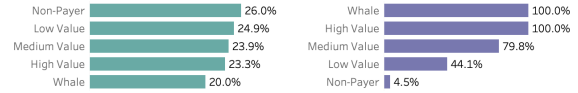
Puzzle Event Dashboard

Funnel Overview

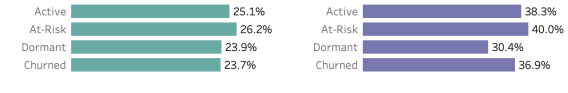
Exposed	Visited	Participated	Completed	Grand Prize
1,800	900 50.0%	750 83.3%	280 37.3%	130 46.4%

Completion is the key engagement bottleneck, with only 37.3% of participants completing the puzzle.

User Value Analysis



Lifecycle Stage Analysis



Monetization Impact

Attributed Payers	Attributed Revenue	ARPPU	Payer Conversion
380	81,312	214	50.7%

50.7% of participants became attributed payers, generating $81,312 in attributed revenue at an ARPPU of $214.

Key Takeaways

1. Completion is the key engagement bottleneck: only 37.3% of participants completed the puzzle.
2. User value predicts engagement depth: completion rises from 4.5% for Non-Payers to 100% for High Value/Whales.

Recommendations

1. Add milestone rewards to reduce participation → completion drop-off.
2. Prioritize Medium-Value users for upsell, given their 79.8% completion rate.
3. Use a control group / A/B test to measure incremental revenue beyond attributed revenue.

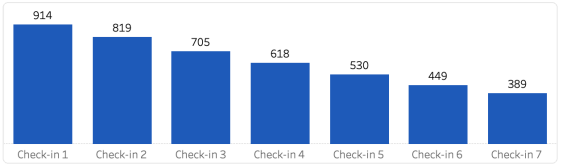
7-Day Check-in Dashboard

Campaign Overview

Exposed	Visited	Unlocked	Completed 7 Check-ins
1,500	1,100 Visit Rate 73.3%	1,037 Unlock Rate 94.3%	389 Completion Rate 37.5%

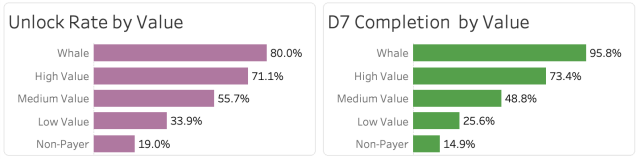
94.3% of visitors unlocked the event, but only 37.5% of unlocked users completed all 7 check-ins.

Check-in Journey



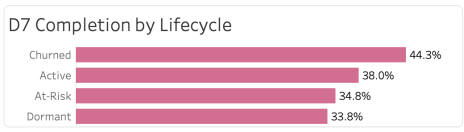
Engagement declined gradually across the journey, with no single sharp drop-off.

User Value Analysis



Engagement and completion increase consistently with user value.

Lifecycle Stage



At-Risk users did not outperform other lifecycle stages, while user value showed much stronger separation.

Takeaways

1. Low unlock friction, but sustained engagement is the challenge: 94.3% of visitors unlocked, but only 37.5% completed all 7 check-ins.
2. User value strongly predicts persistence: D7 completion rises from 14.9% for Non-Payers to 95.8% for Whales.
3. Lifecycle targeting underperformed: At-Risk users completed at 34.8%, below Active users at 38.0%.

Recommodations

1. Improve Exposure → Visit conversion: optimize campaign messaging, placement, and entry points to reduce the 400-user drop-off.
2. Sustain motivation across the journey: use progress reminders and milestone rewards to reduce gradual attrition.
3. Prioritize Medium-Value users for growth: their 48.8% completion rate suggests meaningful engagement with room for uplift.

King's Return Dashboard

Campaign Overview - Did we reactivate dormant users?

Eligible Users	Campaign Exposure	Return Login	Overall Reactivation Rate
687	631	446	64.92%
	Exposure Rate 91.85%	Login Rate 70.68%	

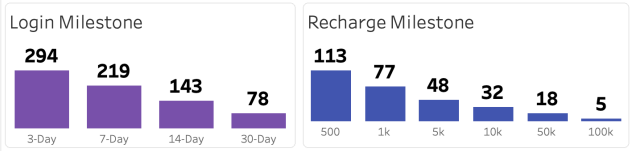
Strong reach and reactivation: 64.92% of eligible dormant users returned during the campaig..

Post-Return Engagement & Retention Journey

Campaign Entry	Gift Claimed	Login 30 Days	30-DAY Retention Rate
401	381	78	20.47%
89.91% of returned user	95.01% of entry users	20.47% of gift claimer	

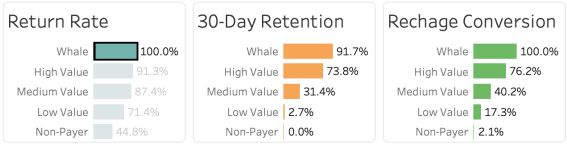
Engagement through entry and gift claim was smooth, but a large drop-off occurred before Day 30.

Milestone Attainment



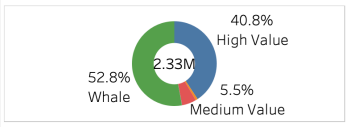
Engagement depth weakened as thresholds increased: fewer users reached longer login milestones or higher recharge tiers.

User Value Tier Performance



Historical user value predicts campaign outcomes: higher-value users returned more, retained longer, and converted to recharge at substantially higher rates.

Revenue Contribution by Value Tier



Revenue was highly concentrated: Whale and High Value users contributed 93.6% of total campaign revenue.

Key Takeaways

1. **Reactivation succeeded:** 64.9% of eligible dormant users returned.
2. **Retention remained the key challenge:** only 20.5% of gift claimers remained active at Day 30.
3. **User value strongly predicted performance:** higher-value users showed stronger return, retention, and recharge behavior.
4. **Revenue was highly concentrated:** Whale and High Value users generated 93.6% of campaign revenue.

Recommendations

1. **Strengthen post-return retention:** add follow-up missions, progressive rewards, and personalized messaging after the initial return gift.
2. **Prioritize high-value win-back:** focus stronger incentives and CRM investment on Whale and High Value dormant users.
3. **Segment incentives by user value:** use lower-friction incentives for lower-value users while reserving premium rewards for historically high-value users.

Fortune Recharge Dashboard

Funnel Overview

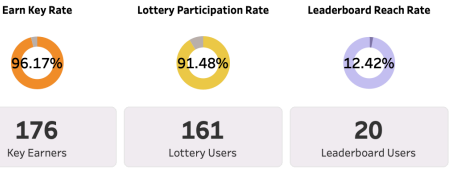
Exposure	Visit	Recharge	Key Earners	Lottery Users	Leaderboard Users
1,800	700	183	176	161	20
	Visit Rate 38.89%	Recharge Rate 26.14%	Key Earn Rate 96.17%	Lottery Draw Rate 91.48%	Leaderboard Rate 12.42%

Recharge is the key monetization bottleneck, with only 26.1% of visitors converting.

Recharge Performance

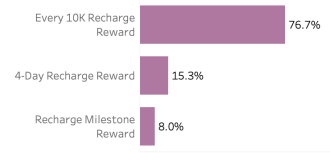
Campaign Users	Recharge Users	Conversion Rate	Total Revenue	ARPPU	Avg Recharge Days
700	183	26.14%	164,294	897.8	4

Key Engagement



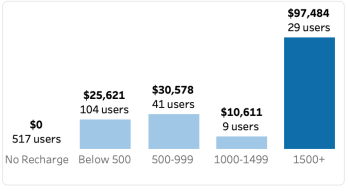
Post-recharge engagement remained high, but only 12.4% of lottery users reached the leaderboard.

Key Distribution



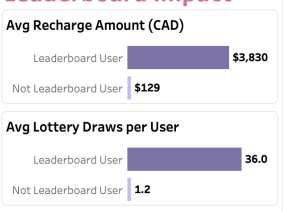
76.7% of earned keys came from the Every 10K Recharge reward, making it the dominant key source.

Revenue by Recharge Tier



1500+ users contributed 59.3% of campaign revenue with only 29 users.

Leaderboard Impact



Leaderboard users recharged ~29.7× more and made 30× more lottery draws than other users.

Key Takawyas

1. **Recharge was the key monetization bottleneck:** only 26.1% of campaign visitors recharged.
2. **Post-recharge engagement was strong:** 96.2% earned keys and 91.5% joined the lottery.
3. **Revenue was highly concentrated:** the 1,500+ tier generated 59.3% of revenue with only 29 users.
4. **Leaderboard users were exceptionally valuable:** $3,830 vs $129 avg recharge and 36.0 vs 1.2 lottery draws.

Recommendations

1. **Improve recharge conversion:** strengthen first-recharge incentives and targeted reminders.
2. **Protect high-value users:** use differentiated rewards and personalized CRM for the 1,500+ segment.
3. **Leverage leaderboard mechanics:** increase leaderboard visibility and test targeted participation incentives.

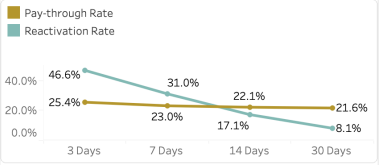
SMS Dashboard

Overview

Total SMS Sent	Delivered	Reactivated (7d)	Paid (7d)	Reactivated Revenue (7d)
6,288	4,539	1,516	370	$178,174
	Delivery Rate 72.2%	Reactivation Rate 33.4%	Pay-through Rate 24.4%	

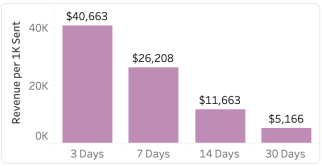
SMS reactivated one-third of delivered users, while nearly one-quarter of reactivated users paid within 7 days.

Reactivation vs Pay-through by Lapse Depth



Reactivation declined sharply with lapse depth, while pay-through remained relatively stable.

Revenue Efficiency by Lapse Depth



Revenue per 1K messages fell from $40.7K at 3 days to $5.2K at 30 days — nearly an 8× decline.

Where to Focus

3 DAYS	7 DAYS	14 DAYS	30 DAYS
46.6% Reactivation	31.0% Reactivation	17.1% Reactivation	8.1% Reactivation
$40.7K Revenue / 1K	$26.2K Revenue / 1K	$11.7K Revenue / 1K	$5.2K Revenue / 1K
PRIORITY	CORE	SELECTIVE	MAINTENANCE

Focus SMS on 3–7 day lapses, where both reactivation and revenue efficiency are strongest.

SMS Content Strategy

Personalized social-proximity hooks

Profile Visitor
"Someone viewed your profile!"

Nearby Interest
"Someone near you is interested!"

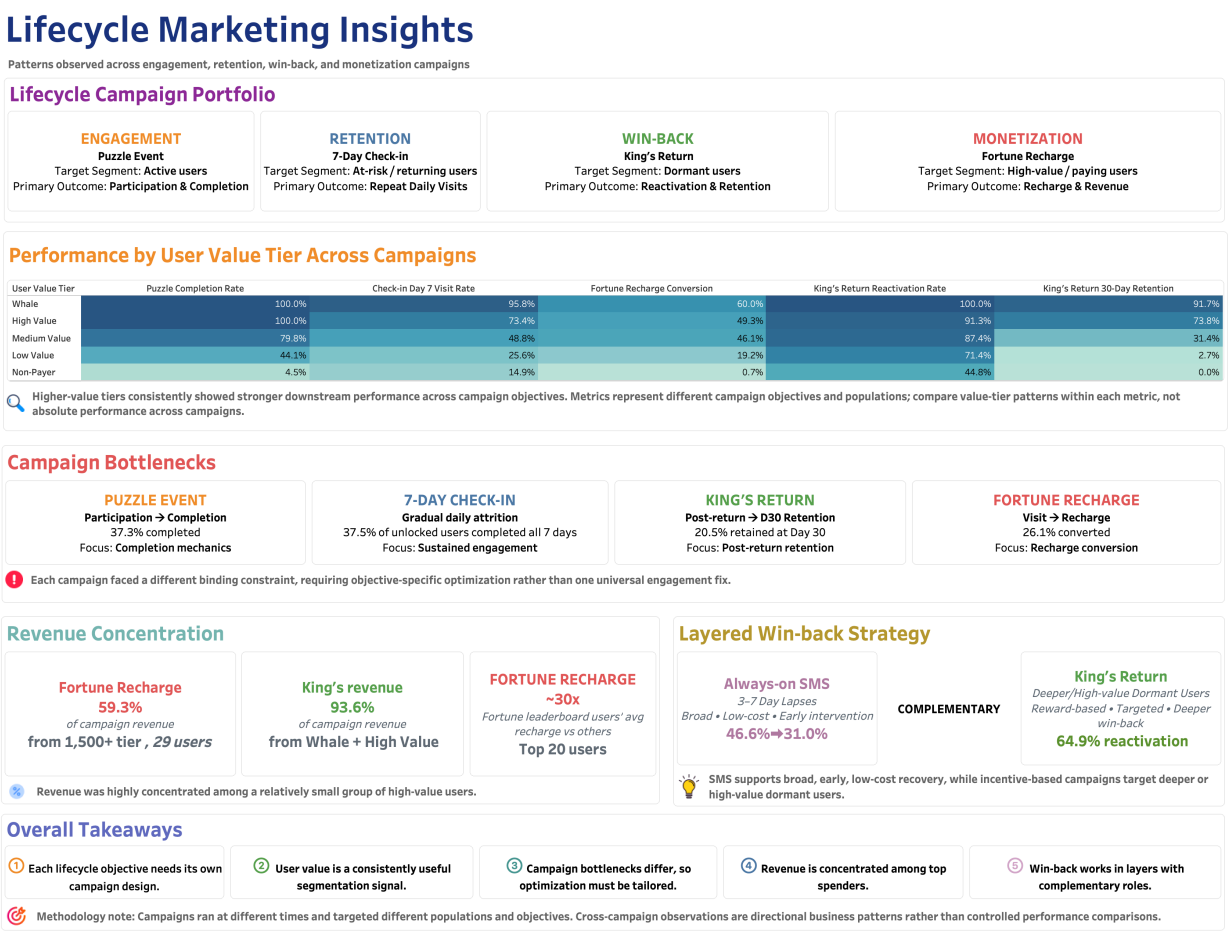
Unread Message
"You have an unread message!"

Key Takeaways

1. **Reactivation declined sharply with lapse depth:** 46.6% of 3-day lapsed users returned, compared with only 8.1% at 30 days.
2. **Pay-through remained relatively stable:** 21.6–25.4% of reactivated users paid within 7 days across lapse depths.
3. **Revenue efficiency fell substantially:** revenue per 1K messages dropped from $40.7K at 3 days to $5.2K at 30 days.
4. **SMS worked best as an early win-back channel:** 3–7 day lapses delivered the strongest combination of reactivation and revenue efficiency.

Recommendations

1. **Prioritize early intervention:** concentrate SMS on 3–7 day lapses, where reactivation and revenue efficiency are strongest.
2. **Layer win-back channels:** use low-cost SMS for shallow lapses and richer reward campaigns for deeper or high-value dormant users.
3. **Add a holdout group:** measure incremental reactivation and revenue before scaling the program.



1. Business Background

A live entertainment and social platform aimed to improve user engagement, retention, reactivation, and monetization through CRM and lifecycle marketing campaigns.

As the platform expanded, several business challenges emerged:

- Some active users gradually became less engaged
- Users with declining activity required retention strategies
- Dormant users stopped returning to the platform
- High-value users required stronger monetization incentives

To address these issues, the operations team launched multiple CRM campaigns aligned to different lifecycle objectives (engagement, retention, win-back, monetization), supported by an always-on SMS program that reactivates lapsed paying users.

This project reorganizes and analyzes those campaigns from a Lifecycle Marketing and CRM Analytics perspective.

2. Business Objective

The main business objectives of this project are:

- Improve user engagement and platform activity
- Increase user retention and return frequency

- Reactivate dormant and lapsed users (through reward campaigns and an always-on SMS program)
- Improve recharge conversion and monetization
- Enhance long-term user value (LTV)
- Optimize CRM campaign performance across different lifecycle stages

3. CRM Targeting Framework

Target Segment	User Status	Main Business Goal	Segment Type
Active Users	Frequently active users	Engagement	Lifecycle stage
At-Risk Users	Users with declining activity	Retention	Lifecycle stage
Dormant Users	Long-term inactive users (14+ days)	Win-back	Lifecycle stage
Churned / Lapsed Users	Longer-lapsed, previously paying users	Reactivation (always-on SMS)	Lifecycle stage
High-Value Users	High recharge / whale users	Monetization	Value tier

While most campaigns were available to all users, each campaign was primarily designed to influence user behavior associated with a specific lifecycle objective. These lifecycle stages served as strategic CRM goals rather than strict campaign audience definitions. Win-back (Dormant) is addressed by a reward campaign (King's Return), while deeper-lapsed users are handled by an always-on SMS reactivation program.

The first four segments are lifecycle stages (defined by activity recency); the last is a value tier (defined by historical spend). The platform aligns each CRM objective to whichever segment best fits it, rather than forcing all four onto a single dimension.

4. Campaign Strategy & KPI Framework

CRM Objective	Campaign Type	Example Campaign	Main Goal	Core KPI
Engagement	Engagement Campaign	Puzzle Event	Increase participation and engagement	Participation Rate
Retention	Retention Campaign	7-Day Check-in Campaign	Improve user retention and return frequency	D7 Completion Rate
Win-back	Win-back Campaign	King's Return Campaign	Reactivate dormant users and improve post-return retention	Reactivation Rate
Reactivation	Always-on Channel	SMS Reactivation Program	Win back lapsed paying users through message content alone	Reactivation Rate · Revenue per Send
Monetization	Monetization Campaign	Fortune Recharge Campaign	Increase recharge frequency and monetization	Revenue / ARPPU

5. CRM Campaign Operations

5.1 Engagement Campaign

Campaign Example: Puzzle Event

Objective: Increase user participation and platform engagement.

Target Audience: All Users.

Campaign Mechanics

- Users receive 1 puzzle attempt from daily activity check-ins.
- Additional puzzle attempts are earned through recharge milestones.
- Each puzzle piece unlocks a random reward from a reward pool.

- Completing all puzzle pieces unlocks a grand prize package.
- Daily participation and recharge behaviors drive puzzle progression.

Main KPI: Participation Rate · Puzzle Completion Rate · Reward Claim Rate · Recharge Conversion Rate · Average Recharge Amount

5.2 Retention Campaign

Campaign Example: 7-Day Check-in Campaign

Objective: Increase user retention and return frequency through a low-cost commitment mechanism and progressive reward system.

Target Audience: All Users

Campaign Mechanics

- Users pay 100 Diamonds once to unlock the campaign.
- Users receive one reward for each completed check-in day.
- Check-ins do not need to be consecutive.
- Users can complete 7 check-ins within a 14-day campaign period.
- The final reward tier unlocks premium cosmetic items and special rewards.

Main KPI: Campaign Activation Rate · Check-in Completion Rate · D7 Completion Rate · Return Visit Rate · Recharge Conversion Rate

5.3 Win-back Campaign

Campaign Example: King’s Return Campaign

Objective: Reactivate dormant users and improve post-return retention through structured reward progression and social re-engagement mechanisms.

Target Audience: Users inactive for 14+ days

Campaign Mechanics

- Automatically triggered when dormant users return to the platform.
- Users receive a temporary King’s Return status tag.
- Users can claim free rewards through a 7-day login progression system.
- Users can unlock a premium reward track after spending 600 Diamonds.
- Rewards must be completed within a 14-day reactivation window.

Main KPI: Reactivation Rate · Return Login Rate · D7 Post-Return Retention · Premium Unlock Rate · Recharge Conversion Rate

5.4 Monetization Campaign

Campaign Example: Fortune Recharge Campaign

Objective: Increase recharge frequency, user spending, and overall platform revenue through layered reward mechanisms and competitive ranking systems.

Target Audience: All Users

Campaign Mechanics

- Users earn reward keys through recharge activities.
- Consecutive recharge days unlock additional rewards.
- Cumulative recharge milestones unlock premium cosmetic rewards.

- Reward keys can be used in a randomized prize machine.
- Prize machine rewards include cosmetic items and ranking points.
- Ranking rewards are distributed based on accumulated ranking points.
- Higher recharge activity generates more reward opportunities.

Main KPI: Recharge Participation Rate · Recharge Conversion Rate · Total Recharge Revenue · ARPPU · Average Recharge Amount · Ranking Participation Rate

5.5 Reactivation Program (Always-on)

Campaign Example: SMS Reactivation Program

Objective: Win back lapsed paying users who have gone inactive, using message content alone rather than an in-app reward mechanic.

Target Audience: Previously paying users (historical recharge above a spend threshold) who have lapsed, segmented by lapse depth (3 / 7 / 14 / 30 days inactive).

Campaign Mechanics

- Lapsed paying users are identified on a rolling basis and contacted by SMS.
- Messages use personalized, social-proximity hooks (a visitor viewed your profile, someone nearby, an unread message) rather than discounts.
- Sends are batched by lapse depth, so reactivation can be measured against how long a user has been gone.
- No reward or in-app mechanic is attached; the program tests content-only reactivation.

Main KPI: Delivery Rate · Reactivation Rate · Pay-through Rate · Revenue per Send

6. CRM Channels

CRM Channel	Purpose
Push Notification	Activity reminder & engagement
SMS	Lapsed / dormant user reactivation
Homepage Banner	Campaign exposure
Splash Screen	High-traffic campaign promotion
Room Entry Icon	In-app activity conversion
Official Notification	Reward reminder & campaign engagement
Activity Center	Centralized campaign distribution

7. Funnel Design

7.1 Puzzle Event Funnel

Business Goal. Increase user engagement and encourage recharge-driven participation through a gamified puzzle reward system.

Primary Funnel

```
Campaign Exposure → Activity Visit → Puzzle Attempt Earned → Puzzle Piece Unlocked → Puzzle Completion → Grand Prize Claim
```

Funnel Stage	Definition
Campaign Exposure	User was exposed to the campaign through banners, splash screens, push notifications, or in-app promotions
Activity Visit	User entered the activity page

Funnel Stage	Definition
Puzzle Attempt Earned	User earned at least one puzzle attempt through check-in or recharge tasks
Puzzle Piece Unlocked	User used an attempt and unlocked at least one puzzle piece
Puzzle Completion	User unlocked all required puzzle pieces
Grand Prize Claim	User claimed the final completion reward

Metric	Formula
Visit Rate	Activity Visitors / Exposed Users
Attempt Rate	Users Earning Attempts / Activity Visitors
Participation Rate	Users Unlocking Puzzle Pieces / Activity Visitors
Completion Rate	Puzzle Completed Users / Puzzle Participants
Grand Prize Claim Rate	Grand Prize Claim Users / Puzzle Completed Users
Recharge Conversion Rate	Recharge Users / Activity Visitors
ARPPU	Total Revenue / Paying Users

Supporting Analysis: Recharge Impact. Evaluate whether recharge-generated attempts contributed to puzzle completion and reward claiming

```
Activity Visit → Recharge → Additional Attempts Earned → Puzzle Completion → Grand Prize Claim
```

Metric	Formula
Recharge Conversion Rate	Recharge Users / Activity Visitors
Revenue	Total recharge amount from activity visitors during the campaign window
ARPPU	Total Revenue / Paying Users
Completion Rate by Recharge Status	Completed Users / Participants
Grand Prize Claim Rate by Recharge Status	Claim Users / Completed Users

7.2 7-Day Check-in Funnel

Business Goal. Improve user retention and return frequency through progressive daily rewards.

Primary Funnel

```
Campaign Exposure → Activity Visit → Unlock → Check-in Progression → Day 7 Completion
```

Funnel Stage	Definition
Campaign Exposure	User was exposed to the campaign
Activity Visit	User entered the campaign page
Unlock	User paid the unlock cost and joined the event
Check-in Progression	User continued daily check-ins throughout the event
Day 7 Completion	User completed all seven check-ins

Metric	Formula
Visit Rate	Activity Visitors / Exposed Users
Unlock Rate	Unlock Users / Activity Visitors
Completion Rate	Day 7 Users / Unlock Users
Revenue per Unlock User	Revenue / Unlock Users

7.3 King’s Return Funnel

Business Goal. Reactivate dormant users and improve post-return retention through return rewards and milestone-based engagement incentives.

Primary Funnel

```
Eligible Dormant Users → Campaign Exposure → Return Login → Campaign Entry → Gift Claimed → 30-Day Login
```

Stage	Definition
Eligible Dormant Users	Users inactive for 14+ days
Campaign Exposure	Dormant users exposed to SMS, Push, campaign banner, pop-up, or activity center entry
Return Login	Dormant users returned and logged in
Campaign Entry	Returned users entered the King’s Return campaign
Gift Claimed	Users claimed the return reward
30-Day Login	Users remained active within 30 days

Metric	Formula
Reactivation Rate	Return Login Users / Eligible Users
Campaign Entry Rate	Campaign Entry Users / Return Login Users
Gift Claim Rate	Gift Claim Users / Campaign Entry Users
30-Day Retention Rate	30-Day Login Users / Campaign Entry Users

Supporting Analysis 1: Login Reward Progression

```
Campaign Entry → 3-Day → 7-Day → 14-Day → 30-Day Reward.
```

Metric	Formula
Claim Rate	Claimed Users / Qualified Users
Reward Progression Rate	Next Milestone Users / Previous Milestone Users

Supporting Analysis 2: Recharge Reward Progression

```
Campaign Entry → Recharge 500 → 1,000 → 5,000 → 10,000 → 50,000 → 100,000.
```

Metric	Formula
Claim Rate	Claimed Users / Qualified Users
Tier Progression Rate	Next Tier Qualified Users / Previous Tier Qualified Users

7.4 Fortune Recharge Funnel

Business Goal. Increase recharge frequency and platform revenue through recharge rewards, fortune keys, lottery participation, and leaderboard competition.

Primary Funnel

```
Campaign Exposure → Activity Visit → Recharge → Key Earned → Lottery Participation → Wealth Coin Earned → Leaderboard Participation
```

Funnel Stage	Definition
Campaign Exposure	User was exposed to the Fortune Recharge campaign
Activity Visit	User entered the campaign page

Funnel Stage	Definition
Recharge	User completed at least one recharge during the campaign
Key Earned	User earned at least one Fortune Key through any campaign reward mechanism
Lottery Participation	User used Fortune Keys in the lottery system
Leaderboard Participation	User appeared on the campaign leaderboard

Metric	Formula
Visit Rate	Activity Visitors / Exposed Users
Recharge Conversion Rate	Recharge Users / Activity Visitors
Key Earn Rate	Key Earned Users / Recharge Users
Lottery Participation Rate	Lottery Users / Key Earned Users
Leaderboard Participation Rate	Leaderboard Users / Lottery Users
ARPPU	Total Revenue / Paying Users

Supporting Analysis 1: Key Distribution Analysis

```
4-Day Recharge Reward vs. Every 10K Recharge Reward vs.Recharge Milestone Reward
```

Supporting Metrics 1

Metric	Formula
Key Distribution	Keys Generated by Reward Type / Total Keys

Supporting Analysis 2: Reward Mechanism Analysis

```
Key Earned → Lottery Participation → Leaderboard Participation
```

Supporting Metrics 2

Metric	Formula
Lottery Participation Rate	Lottery Users / Key Users
Leaderboard Participation Rate	Leaderboard Users / Lottery Users
Average Lottery Draws per User	Total Lottery Draws / Lottery Users

Supporting Analysis 3: User Value Analysis

```
Whale vs. High Value vs. Medium Value vs. Low Value vs. Non-Payer
```

Supporting Metrics 3

Metric	Formula
Revenue Contribution	Revenue / Total Revenue
ARPPU	Total Revenue / Paying Users
Average Recharge Days	Total Recharge Days / Paying Users

Supporting Analysis 4: Leaderboard Analysis

```
Leaderboard Users vs. Non-Leaderboard Users
```

Supporting Metrics 4

Metric	Formula
Average Recharge Amount	Revenue / Users
Average Lottery Draws	Total Lottery Draws / Users
Average Wealth Coins	Total Wealth Coins / Users

8. Data Preparation & Modeling

Before any analysis, raw campaign data was validated and standardized so that metrics, segmentation, and funnel logic stay consistent across every CRM campaign, the SQL analysis, and the Tableau dashboard.

This project is based on real CRM campaigns. Because the original production data could not be retained, the dataset was synthetically reconstructed to reproduce the analysis and demonstrate the methodology. Validation therefore focuses on cross-table consistency and business-rule integrity rather than raw data cleansing.

8.1 Data Preparation & Validation

Purpose: Confirm that every table is internally consistent and safe to aggregate, and that the same metric means the same thing across campaigns.

Validation checks

Check	Purpose
Standardize campaign names	Ensure consistent campaign grouping across `campaign_exposure`, `recharge_log`, and the per-campaign logs
Binary flags contain only 0 / 1	Confirm status fields (`visit_flag`, `recharge_flag`, `leaderboard_flag`, login/claim flags, …) are clean before they drive funnel counts
Recharge amounts ≥ 0	Prevent negative or invalid revenue from distorting totals and ARPPU
Events fall within the campaign window	Confirm exposures and recharges occur inside each campaign's run dates
Funnel monotonicity	A later funnel stage never exceeds an earlier one (e.g. Visit ≤ Exposure, Completion ≤ Participation)
Reward logic	Reward-claim flags never exceed the matching qualification flags (King's login & recharge rewards)
Leaderboard logic	`leaderboard_rank` exists only when `leaderboard_flag = 1`
Referential integrity	Every behavior table joins back to its profile table through `user_id`
Revenue source per campaign	Revenue is taken from each campaign's own log so definitions stay consistent across tables

8.2 Business Modeling

Purpose. Standardize the business rules *before* analysis so that segmentation, funnel metrics, and monetization calculations are identical across SQL and Tableau. Each rule maps to a concrete field so it is fully reproducible.

The platform maintains a tag system that defines user segments through explicit rule-based logic. Two of those segmentations drive this analysis — **User Value Tier** and **Lifecycle Stage**. In this dataset both arrive as pre-computed snapshot fields (`user_value_tier`, `lifecycle_stage`); the underlying rules are summarized below so the segmentation stays transparent and reproducible. Two further segments (Recharge Tier, Campaign Status) are derived directly from event fields during analysis.

User Value Tier

Defined by historical (trailing-12-month) cumulative recharge. *Thresholds below are illustrative — replace with the platform's actual tag-system values.*

Tier	Rule (historical recharge, trailing 12 months)
Whale	≥ 100,000
High Value	20,000 – 100,000

Tier	Rule (historical recharge, trailing 12 months)
Medium Value	2,000 – 20,000
Low Value	1 – 2,000
Non-Payer	0 (never recharged)

In this dataset the tier is provided as a snapshot field (`user_value_tier`) and treated as a stable user trait; because it is a single snapshot, it is applied consistently across the non-concurrent campaigns rather than recomputed per campaign.

Lifecycle Stage

Defined by recent login / activity recency. *Thresholds illustrative — replace with the platform's actual values.*

Stage	Rule
Active	Logged in on ≥ 7 of the last 15 days
At-Risk	Logged in on 1–6 of the last 15 days (declining activity)
Dormant	No activity for 14+ days
Churned	No activity for 30+ days

Also provided as a snapshot field (`lifecycle_stage`) and treated the same way. Values in the data: Active, At-Risk, Dormant, Churned.

Recharge Tier (in-campaign spend)

Derived in analysis from in-campaign spend. Field: `campaign_recharge_amount`.

Tier	Logic
No Recharge	`= 0`
Below 500	`1–499`
500–999	`500–999`
1,000–1,499	`1,000–1,499`
1,500+	`≥ 1,500`

(The King's Return revenue analysis uses its own wider spend bands — 500–999, 1k–5k, ..., 100k+ — because that campaign's reward tiers run much higher.)

Campaign Status (funnel flags)

We use Fortune as the worked example because it has the longest funnel and richest field set; the same modeling approach (stage = queryable flag, Raw→Model→Analysis) applies to all four campaigns, whose stage definitions appear in their respective sections

Status	Logic
Exposed	Has a record in `campaign_exposure`
Visited	`visit_flag = 1`
Recharged	`recharge_flag = 1`
Key Earned	`key_earned_total > 0`
Lottery Participant	`lottery_draw_count > 0`
Leaderboard User	`leaderboard_flag = 1`

8.3 Analytics Workflow

Purpose. Business rules are applied *before* SQL analysis so that KPI calculations, user segmentation, and dashboard metrics stay consistent throughout the project.

```
Raw Campaign Tables
        |
        ▼
Data Preparation & Validation
        |
        ▼
Business Modeling
(User Value Tier · Recharge Tier · Campaign Status)
        |
        ▼
SQL Analysis
        |
        ▼
Tableau Dashboard
        |
        ▼
Business Insights
```

8.4 Analytics Data Model

The model is split into two user domains joined on `user_id`. The general domain covers Puzzle, Check-in, and Fortune; the King's Return domain is kept separate because it targets a specific dormant population that does not overlap the general user base.

```
GENERAL CAMPAIGN DOMAIN
   user_profile  (1 row / user)
        |  user_id
        ├── campaign_exposure      (1 row / exposure event)
        ├── recharge_log           (1 row / recharge transaction)
        ├── puzzle_event_log       (1 row / participant)
        ├── checkin_event_log      (1 row / participant)
        └── fortune_recharge_log   (1 row / visitor)

KING'S RETURN DOMAIN
   kings_user_profile  (1 row / eligible user)
        |  user_id
        ├── kings_campaign_exposure              (1 row / exposure event)
        ├── kings_return_log_with_login_claims   (1 row / eligible user)
        └── kings_recharge_log                   (1 row / recharge transaction)
```

Table grain

Table	Grain	Main analysis supported
`user_profile`	One record per user	Segmentation, lifecycle, user-value analysis
`campaign_exposure`	One record per exposure event	Exposure, visit rate, channel performance
`recharge_log`	One record per recharge transaction	Puzzle revenue, recharge conversion, ARPPU
`puzzle_event_log`	One record per participating user	Puzzle funnel and engagement
`checkin_event_log`	One record per participating user	Retention and check-in completion
`fortune_recharge_log`	One record per participating (visitor) user	Monetization, reward mechanism, leaderboard
`kings_user_profile`	One record per eligible dormant user	Dormant-user segmentation
`kings_campaign_exposure`	One record per exposure event	Win-back exposure performance
`kings_return_log_with_login_claims`	One record per eligible dormant user	Reactivation funnel, reward claims, 30-day retention
`kings_recharge_log`	One record per recharge transaction	Win-back recharge revenue

8.5 Table Dictionary

8.5.1 user_profile

Column Name	Description
`user_id`	Unique user identifier
`register_date`	User registration date
`device_type`	User device type
`total_recharge`	Historical cumulative recharge amount
`user_value_tier`	User value segment based on historical recharge
`inactive_days_current`	Number of inactive days before the current lifecycle snapshot
`lifecycle_stage`	User lifecycle status: **Active, At-Risk, Dormant, or Churned** (note: "High-Value" is a `user_value_tier`, not a lifecycle stage)
`vip_level`	User VIP level
`recharge_count_3m`	Number of recharge transactions in the last 3 months
`recharge_amount_3m`	Recharge amount in the last 3 months
`campaign_participation_count`	Number of campaigns the user participated in
`campaign_completion_count`	Number of campaigns the user completed
`engagement_segment`	Engagement classification based on recent activity

8.5.2 campaign_exposure

Column Name	Description
`exposure_id`	Unique exposure record ID
`user_id`	User identifier
`campaign_name`	Campaign name
`campaign_objective`	Campaign objective: Engagement, Retention, Win-back, or Monetization
`channel`	Campaign delivery channel
`exposure_time`	Time when the user was exposed to the campaign
`click_flag`	Whether the user clicked the campaign entry
`click_time`	Time when the user clicked the campaign entry
`visit_flag`	Whether the user entered the campaign page

8.5.3 recharge_log

Column Name	Description
`recharge_id`	Unique recharge transaction ID
`user_id`	User identifier
`campaign_name`	Campaign associated with the recharge
`recharge_time`	Recharge timestamp
`recharge_amount`	Recharge amount

8.5.4 puzzle_event_log

Column Name	Description
`user_id`	User identifier
`daily_checkin_flag`	Whether the user earned puzzle chances through daily check-in
`recharge_chance_count`	Number of puzzle chances earned through recharge
`chance_count`	Total puzzle chances earned
`puzzle_piece_count`	Number of puzzle pieces unlocked
`completed_flag`	Whether the user completed the full puzzle

Column Name	Description
random_reward_claim_count	Number of random rewards claimed
grand_prize_claim_flag	Whether the user claimed the grand prize

8.5.5 checkin_event_log

Column Name	Description
user_id	User identifier
unlock_flag	Whether the user unlocked the check-in campaign
unlock_cost	Cost paid to unlock the event
total_checkin_days	Total number of completed check-in days
max_checkin_day	Highest check-in count reached (= total_checkin_days in this data, since counts accrue without gaps).
day1_flag ... day7_flag	Whether the user reached the Nth check-in (1st–7th). 'Day N' = the Nth check-in count, based on H5 page entries, NOT calendar day.
final_reward_claim_flag	Whether the user claimed the final reward

8.5.6 fortune_recharge_log

Column Name	Description
user_id	User identifier
visit_flag	Whether the user visited the Fortune Recharge campaign page
recharge_day_count	Number of recharge days during the campaign
campaign_recharge_amount	Total recharge amount during the campaign
key_earned_4day	Keys earned from the 4-day recharge reward
key_earned_10k	Keys earned from the repeated 10K recharge reward
key_earned_milestone	Keys earned from recharge milestone rewards
key_earned_total	Total keys earned
lottery_draw_count	Number of lottery draws completed
wealth_coin	Wealth coins earned from lottery draws
leaderboard_flag	Whether the user entered the leaderboard
leaderboard_rank	User leaderboard rank
recharge_flag	Whether the user made a recharge during the campaign
tier_50k_flag ... tier_1000k_flag	Whether the user reached each recharge milestone (50K / 100K / 500K / 1000K)

8.5.7 kings_user_profile

Column Name	Description
user_id	King's Return user identifier
user_value_tier	Historical user value tier before reactivation
historical_recharge	Historical cumulative recharge amount
vip_level	User VIP level
registration_days	Number of days since registration
inactive_days_before_campaign	Inactive days before campaign eligibility

8.5.8 kings_campaign_exposure

Column Name	Description
exposure_id	Unique exposure record ID

Column Name	Description
`user_id`	King's Return user identifier
`campaign_name`	Campaign name
`campaign_objective`	Campaign objective
`channel`	Campaign delivery channel
`exposure_time`	Exposure timestamp
`click_flag`	Whether the user clicked the campaign
`click_time`	Click timestamp
`visit_flag`	Whether the user visited the campaign page

8.5.9 kings_return_log_with_login_claims

Column Name	Description
`user_id`	King's Return user identifier
`eligible_flag`	Whether the user was eligible for the campaign
`return_login_flag`	Whether the dormant user logged back in during the reactivation window
`campaign_entry_flag`	Whether the user entered the campaign page
`first_return_gift_received_flag`	Whether the user received the first return gift
`login_3day_flag` ... `login_30day_flag`	Whether the user reached each login milestone (3 / 7 / 14 / 30 days)
`claim_login_3day_reward_flag` ... `claim_login_30day_reward_flag`	Whether the user claimed each login reward (3 / 7 / 14 / 30 days)
`recharge_500_qualified_flag` ... `recharge_100000_qualified_flag`	Whether the user qualified for each recharge reward tier (500 → 100,000)
`claim_recharge_500_reward_flag` ... `claim_recharge_100000_reward_flag`	Whether the user claimed each recharge reward (500 → 100,000)

8.5.10 kings_recharge_log

Column Name	Description
`recharge_id`	Unique recharge transaction ID
`user_id`	King's Return user identifier
`campaign_name`	Campaign name
`recharge_time`	Recharge timestamp
`recharge_amount`	Recharge amount

8.6 Standard Metric Definitions

Metric	Definition
Visit Rate	Visit Users / Exposed Users
Recharge Conversion Rate	Recharge Users / Campaign Visitors
ARPPU	Total Revenue / Paying Users
Participation Rate	Participating Users / Campaign Visitors
Completion Rate	Completed Users / Participating Users（users who performed the campaign's core participation action (e.g. unlocking a puzzle piece in Puzzle, activating the event in 7-Day Check-in, visiting the activity in Fortune Recharge, and entering the campaign in King's Return).）
Claim Rate	Claimed Users / Qualified Users
Key Earn Rate	Key Users / Recharge Users

Metric	Definition
Lottery Participation Rate	Lottery Users / Key Users
Leaderboard Participation Rate	Leaderboard Users / Lottery Users
Return Rate	Return Login Users / Eligible Users
30-Day Retention Rate	30-Day Login Users / Campaign Entry Users

9. Cross-Campaign Overview

The four campaigns are **not concurrent**: Puzzle (Feb 17–24) · Check-in (May 19–26) · Fortune (Jun 16–23) · King's (Sep 1–14). The three general campaigns draw from a shared 3,000-user pool; King's is a separate 687-user dormant pool. This table is therefore a side-by-side reference, not a same-period comparison.

Campaign	Objective	Window (2024)	Exposed/Eligible	Visited/Returned	Participated/Entered	Recharge Users	Reven
Puzzle Event	Engagement	Feb 17–24	1,800	900	750	380	81,312
7-Day Check-in	Retention	May 19–26	1,500	1,100	1,037	—	—
Fortune Recharge	Monetization	Jun 16–23	1,800	700	183	183	164,29
King's Return	Win-back	Sep 1–14	687	446	401	113	2,331,1

Findings

1. **The four campaigns play very different business roles.** King's Return earned the most (2,331,188, from winning back high-value dormant users), Fortune Recharge next (164,294), and Puzzle (81,312) is engagement with incidental monetization; Check-in earns nothing directly. They should not be judged on one yardstick.
2. **Each campaign loses users at a different point.** Fortune loses most people at the very top (1,800 exposed but only 700 visited, 39%), while Check-in held onto far more (1,500 → 1,100, 73%). The part of the funnel that needs fixing differs by campaign.
3. **King's is the smallest pool with the highest value per user.** Only 687 users were eligible, yet it produced the highest revenue of any campaign — a "small pool, high yield" win-back whose return depends on a few reactivated whales.
4. **The table is a reference, not a same-period comparison.** Because the campaigns ran at different times and reached different audiences, any cross-campaign rollup must watch the metric definitions, especially revenue (see notes 1 and 2 and §8.7).

10. Puzzle Event Analysis

10.1 Funnel Analysis

Questions

- How many users visited the activity page?
- What was the overall participation rate?
- What was the puzzle completion rate?
- What percentage of users claimed the grand prize?
- Where did the largest drop-off occur in the event funnel?

Results

Stage	Users	Conversion Rate	Drop-off Users
Exposure	1,800	-	-
Visit	900	50.00%	900
Participation	750	83.33%	150
Completion	280	37.33%	470
Grand Prize Claim	130	46.43%	150

Findings

1. **Visitors participated readily.** Among users who entered the activity page, 83.33% earned a puzzle piece and became participants.
2. **The largest drop-off was Participation → Completion.** Only 37.33% of participants completed the puzzle, so the mechanics required sustained effort that most users did not finish.
3. **Rewards motivated the final step.** Among users who completed the puzzle, 46.43% went on to claim the grand prize, showing the reward was a strong pull once completion was in reach.

10.2 Recharge Impact

Questions

- How many participants made a recharge during the event?
- What was the recharge conversion rate?
- How much revenue was generated?
- What was the average revenue per paying user (ARPPU)?
- To what extent did the event drive monetization?

Results

Metric	Result
Participants	750
Recharge Users	380
Recharge Conversion Rate	50.67%
Total Revenue	81,312
ARPPU	213.98

Findings

1. **Half of participants paid.** The event converted 50.67% of participants into paying users.
2. **Revenue was meaningful, not minimal.** The campaign generated 81,312 in recharge revenue at an ARPPU of 213.98, so paying users contributed real value rather than token purchases.
3. **Engagement and revenue were balanced.** The event drove participation while still monetizing, creating value for both users and the business.

10.3 User Value Analysis

Questions

- Which user segments were most likely to participate?
- Which user segments achieved the highest completion rates?
- How did participation behavior differ across value tiers?

- Did high-value users engage more deeply with the activity?
- Which user groups should be prioritized in future campaigns?

Results

Tier Participation Rate by User Value Tier

User Value Tier	Participants	Total Users	Participation Rate
Non-Payer	312	1,200	26.00%
Low Value	261	1,050	24.86%
Medium Value	129	540	23.89%
High Value	42	180	23.33%
Whale	6	30	20.00%

Completion Rate by User Value Tier

User Value Tier	Participants	Completed Users	Completion Rate
Whale	6	6	100.00%
High Value	42	42	100.00%
Medium Value	129	103	79.84%
Low Value	261	115	44.06%
Non-Payer	312	14	4.49%

Findings

1. **Participation was even across tiers.** Rates sat in a narrow 20%–26% band, so the event appealed broadly across the user base.
2. **Completion rose sharply with value.** All Whale and High Value participants completed the puzzle, versus only 4.49% of Non-Payers.
3. **Value predicts engagement depth.** Higher-value users committed to finishing the event, making them the most valuable audience for future monetization-focused campaigns.

10.4 Lifecycle Stage Analysis

Questions

- Did participation rates differ by lifecycle stage? Was the campaign targeted by stage?
- Did completion rates differ by stage?

Results

lifecycle_stage	Stage Users	Participation Rate	Completion Rate
Active	1,667	25.1%	38.3%
At-Risk	591	26.2%	40.0%
Dormant	468	23.9%	30.4%
Churned	274	23.7%	36.9%

Findings

1. **Participation was flat across stages.** Rates stayed within 23.7%–26.2%, indicating the campaign was broadcast rather than aimed at any particular stage.
2. **Completion was also close across stages.** Rates ran 30%–40%, unlike the value-tier split in §10.3 (100% down to 4.5%). For this campaign, how much a user spends predicts depth far more than their current activity level.

10.5 Key Takeaways

1. **The event attracted participation.** It converted 83.33% of visitors into active participants.
2. **The event showed strong monetization among participants.** 50.67% of participants became attributed payers, generating 81,312 in attributed recharge revenue. However, without a control group, this should not be interpreted as incremental campaign impact.
3. **Appeal was broad.** Participation was consistent across user value segments, ranging from 20% to 26%.
4. **Depth scaled strongly with user value.** Completion increased dramatically with user value, from 4.49% among Non-Payers to 100% among High-Value and Whale users.
5. **Lifecycle stage was less predictive of engagement depth.** Participation and completion varied relatively little across lifecycle stages compared with the much stronger differences observed across user value tiers.

10.6 Recommendations

1. Increase completion through milestone rewards.

The largest drop-off occurred between participation and completion, with only 37.33% of participants completing the puzzle. Intermediate rewards at key milestones could encourage continued progression.

2. Focus on Medium-Value users for upsell.

Medium-Value users achieved a 79.84% completion rate, indicating strong engagement. Targeted recharge incentives could help convert this engagement into higher user value.

3. Improve causal measurement with an A/B test.

Current recharge results reflect **attributed revenue** from campaign-linked recharge activity, but do not establish incremental impact. Future campaigns should include a control group or A/B test to estimate the campaign's true incremental effect on recharge conversion and revenue.

11. 7-Day Check-in Analysis

11.1 Funnel Analysis

Questions

- How many users were exposed to the campaign?
- How many users visited the activity page?
- How many users unlocked the check-in event?
- How did participation change throughout the 7-day journey?
- Where did the largest drop-off occur in the funnel?

Results

Funnel Stage	Users	Conversion Rate	Drop-off Users
Exposure	1,500	-	-
Visit	1,100	73.33%	400
Unlock	1,037	94.27%	63
Day 1	914	88.14%	123
Day 2	819	89.61%	95
Day 3	705	86.08%	114
Day 4	618	87.66%	87
Day 5	530	85.76%	88

Funnel Stage	Users	Conversion Rate	Drop-off Users
Day 6	449	84.72%	81
Day 7 / Final Reward	389	86.64%	60

Findings

1. **Visit interest was strong.** 73.33% of exposed users visited the activity page.
2. **The unlock cost was low friction.** 94.27% of visitors unlocked the event, so the 100-Diamond entry cost rarely deterred users.
3. **The largest drop-off was Exposure → Visit.** 400 users did not enter the activity page after seeing the campaign, pointing to an opportunity in campaign visibility, messaging, or entry-point design.
4. **Engagement declined gradually after unlock.** The biggest post-unlock drop was at Day 1, where 123 users did not continue past their first check-in.
5. **Many still finished.** Despite the long cycle, 389 users completed all seven check-ins, showing meaningful long-term engagement.

11.2 User Value Analysis – Unlock Rate

Questions

- Which user segments were most likely to unlock the event?
- Did willingness to participate vary across value tiers?

Results

User Value Tier	Unlock Users	Total Users	Unlock Rate
Whale	24	30	80.00%
High Value	128	180	71.11%
Medium Value	301	540	55.74%
Low Value	356	1,050	33.90%
Non-Payer	228	1,200	19.00%

Findings

1. **Unlock rate rose with value.** Willingness to pay the entry cost increased steadily from Non-Payer to Whale.
2. **Paying users led participation.** Whale and High Value unlock rates were above 70%.
3. **The entry cost deterred low-value users.** Non-Payers unlocked at just 19%, so the 100-Diamond requirement was a real barrier for them.

11.3 User Value Analysis – Day 7 Completion Rate

Questions

- Which user segments were most likely to complete all seven check-ins?
- How did completion behavior differ across value tiers?

Results

User Value Tier	Unlock Users	Day 7 Users	Completion Rate
Whale	24	23	95.83%
High Value	128	94	73.44%

User Value Tier	Unlock Users	Day 7 Users	Completion Rate
Medium Value	301	147	48.84%
Low Value	356	91	25.56%
Non-Payer	228	34	14.91%

Findings

1. **Completion rose with value.** Day-7 completion increased steadily across tiers.
2. **Whales nearly all finished** the full seven-day journey (95.83%).
3. **High Value users were committed too,** completing above 70%.
4. **Non-Payers dropped off early,** completing at just 14.91%.
5. **Value tracked retention** throughout the activity lifecycle.

11.4 User Value Analysis – Average Check-in Days

Questions

- How long did users remain engaged with the event on average?
- Did higher-value users stay active longer?

Results

User Value Tier	Unlock Users	Avg Check-in Days
Whale	24	6.71
High Value	128	5.95
Medium Value	301	4.72
Low Value	356	3.79
Non-Payer	228	3.22

Findings

1. **Higher-value users stayed engaged longer** than lower-value users on average.
2. **Whales nearly finished the whole journey,** averaging 6.71 check-in days.
3. **Medium Value users were a growth opportunity,** maintaining moderate engagement.
4. **Non-Payers disengaged earliest,** averaging only 3.22 check-in days.
5. **Value is closely linked to long-term participation** and activity retention.

11.5 Lifecycle Stage Analysis

Questions

- Did participation rates differ across lifecycle stages? Did this retention campaign reach its target (At-Risk) stage?
- Did D7 completion differ by stage, and how does it compare to the value-tier cut?

Results

lifecycle_stage	Stage Users	Participation Rate	D7 Completion Rate
Active	1,667	34.7%	38.0%
At-Risk	591	33.5%	34.8%
Dormant	468	31.0%	33.8%

lifecycle_stage	Stage Users	Participation Rate	D7 Completion Rate
Churned	274	42.0%	44.3%

Findings

1. **At-Risk users did not show stronger participation or completion.** Their participation rate was 33.5%, below Active users (34.7%) and Churned users (42.0%). However, because the campaign was open to all users, these results alone cannot establish whether campaign delivery was appropriately targeted.
2. **Stage barely separates completion; value does.** D7 completion ran 34%–44% across stages, whereas by value tier it ranges from Whale 95.8% to Non-Payer 14.9% (§11.3) — value tier separates depth far more sharply than stage.

11.6 Key Takeaways

1. **Exposure converted into participation,** with 73.33% of exposed users visiting and 94.27% of visitors unlocking.
2. **Engagement declined gradually** across the seven-day journey, the core challenge of sustaining long participation.
3. **The biggest loss was Exposure → Visit,** pointing to campaign-entry effectiveness as the main lever.
4. **Higher-value users engaged more,** with higher completion and longer participation.

11.7 Recommendations

1. Improve Exposure-to-Visit conversion. The largest drop-off came before users entered the activity page; optimize banner design, push copy, and placement.

2. Sustain motivation across the whole check-in window. Day-over-day conversion held steadily between 85% and 90% with no single sharp cliff, so attrition was gradual and continuous rather than concentrated on one day. This points to ongoing, every-step encouragement (e.g. small daily streak bonuses) rather than a one-off reward placed on a specific day.

3. Focus on Medium Value growth. These users showed solid participation and moderate completion; targeted incentives could move them up.

4. Create tiered reward structures. High Value and Whale users showed strong sustained engagement; test VIP-exclusive rewards and bonus milestones to extend participation across the campaign journey.

12. King’s Return Analysis

12.1 CRM Reactivation Funnel Analysis

Questions

- How many dormant users returned after receiving the campaign?
- How many users entered the campaign?
- How many users claimed the return gift?
- What was the long-term retention outcome?
- Where did the largest drop-off occur in the reactivation journey?

Results

Stage	Users	Conversion Rate	Drop-off Users
Eligible Users	687	-	-
Campaign Exposure	631	91.85%	56
Return Login	446	70.68%	185
Campaign Entry	401	89.91%	45

Stage	Users	Conversion Rate	Drop-off Users
Gift Claimed	381	95.01%	20
Login 30 Days	78	20.47%	303

KPI	Result
Eligible Users	687
Reactivated Users	446
Overall Reactivation Rate	64.92%

Findings

1. **CRM reach was high.** Of 687 eligible dormant users, 631 were exposed (91.85%), so the channels reached most of the target pool.
2. **Initial reactivation was strong.** 446 users returned and logged in — a 70.68% exposure-to-return conversion and a 64.92% overall reactivation rate across all eligible users — showing the message and return incentive drove return behavior.
3. **Post-return engagement was smooth.** 89.91% entered the campaign and 95.01% claimed the return gift, so the page and reward design created little friction.
4. **The largest drop-off was Gift Claimed → 30-Day Login.** Only 20.47% of gift claimers were still active after 30 days, so short-term incentives drove return but not lasting retention.
5. **Reach and reactivation succeeded; retention did not.** The campaign needs additional post-return retention mechanisms beyond the reward-claim stage.

12.2 Login Reward Claim Analysis

Questions

- How many users qualified for each login milestone reward?
- What percentage of qualified users claimed rewards?
- Did reward engagement decline over time?
- Which reward milestone achieved the strongest participation?

Results

Reward Stage	Qualified Users	Claimed Users	Claim Rate
3 Day Reward	294	265	90.14%
7 Day Reward	219	194	88.58%
14 Day Reward	143	122	85.31%
30 Day Reward	78	65	83.33%

Findings

1. **Claim rates stayed high,** above 83% at every milestone.
2. **Participation tapered with longer requirements,** as expected in retention campaigns.
3. **The 3-Day reward led (90.14%),** while the 30-Day reward held a strong 83.33% despite far greater commitment.
4. **Milestone rewards kept users engaged** after reactivation.

12.3 Recharge Reward Claim Analysis

Questions

- How many users qualified for recharge rewards?
- What percentage of qualified users claimed rewards?
- Did claim behavior vary across recharge tiers?
- How effective were recharge incentives at driving engagement?

Results

Recharge Tier	Qualified Users	Claimed Users	Claim Rate
Recharge 500	113	99	87.61%
Recharge 1,000	77	69	89.61%
Recharge 5,000	48	42	87.50%
Recharge 10,000	32	28	87.50%
Recharge 50,000	18	15	83.33%
Recharge 100,000	5	4	80.00%

Findings

1. **Claim rates were strong at every tier,** consistently above 80%.
2. **High thresholds did not dampen claiming,** so users were motivated to collect once they qualified.
3. **The 1,000 tier led (89.61%),** and even the 100,000 tier held at 80.00%.
4. **Recharge incentives reinforced spending** and encouraged users to maximize purchase value.

12.4 User Value Analysis

Questions

- Which user segments were most likely to return?
- Which segments achieved the strongest long-term retention?
- Which segments generated the highest monetization outcomes?
- How did reactivation effectiveness vary by user value tier?
- Which user groups should be prioritized in future campaigns?

Return Rate by User Value Tier

User Value Tier	Eligible Users	Return Users	Return Rate
Whale	12	12	100.00%
High Value	46	42	91.30%
Medium Value	119	104	87.39%
Low Value	224	160	71.43%
Non-Payer	286	128	44.76%

30-Day Retention by User Value Tier

User Value Tier	Campaign Entry Users	Login 30-Day Users	Retention Rate
Whale	12	11	91.67%
High Value	42	31	73.81%
Medium Value	102	32	31.37%
Low Value	150	4	2.67%
Non-Payer	95	0	0.00%

Recharge Conversion by User Value Tier

User Value Tier	Campaign Entry Users	Recharge Users	Conversion Rate
Whale	12	12	100.00%
High Value	42	32	76.19%
Medium Value	102	41	40.20%
Low Value	150	26	17.33%
Non-Payer	95	2	2.11%

Note: “Recharge Users” is counted via `recharge_500_qualified_flag`, which equals the number of users with a record in `kings_recharge_log` (the same here, since no one recharged below 500). King’s has no `lifecycle_stage` column and its users do not overlap the general pool, so it is not included in the lifecycle-stage analysis; use `inactive_days_before_campaign` for a “dormancy depth” cut instead.

Findings

1. **Reactivation rose with historical value.** Return, retention, and recharge conversion all increased with the user’s prior value tier.
2. **Whale and High Value led every metric,** with the highest return, retention, and recharge conversion rates.
3. **Non-Payers were a large but weak segment,** accounting for many eligible users but very low long-term retention and monetization.
4. **Historical value predicts post-reactivation behavior,** making it a strong targeting signal.
5. **High-value users drove the impact** and should remain the primary audience for future reactivation campaigns.

12.5 Recharge Revenue Analysis

Questions

- How much recharge revenue did King’s Return generate? What was the conversion rate?
- How was revenue distributed across spend bands and value tiers? How concentrated was it?

Results

Overall

Metric	Value
Total recharge revenue	2,331,188
Recharge users	113
Recharge conversion (rechargers ÷ campaign entrants)	28.2% (113 / 401)
Mean ARPPU	20,630
Median ARPPU	**3,943**
Top-10 spender share	49.1%

By spend band

Spend Band	Users	Revenue	Contribution	Avg/User
500–999	36	27,683	1.2%	769
1,000–4,999	29	88,338	3.8%	3,046
5,000–9,999	16	123,361	5.3%	7,710
10,000–49,999	14	381,979	16.4%	27,284
50,000–99,999	13	1,025,104	44.0%	78,854
100,000+	5	684,723	29.4%	136,945

By user value tier

User Value Tier	Rechargers	Revenue	Contribution	Mean ARPPU	Median ARPPU
Whale	12	1,231,754	52.8%	102,646	89,970
High Value	32	950,015	40.8%	29,688	20,105
Medium Value	41	128,429	5.5%	3,132	2,811
Low Value	26	19,387	0.8%	746	756
Non-Payer	2	1,603	0.1%	802	802

Findings

1. **Revenue is highly concentrated.** The two bands at 50,000 and above are just 18 users but 73.4% of revenue; by value tier, Whale and High Value together are 93.6%.
2. **The mean is inflated by whales.** Mean ARPPU (20,630) far exceeds the median (3,943), so report both and lead with the median for the typical payer.
3. **The high revenue is by design, not a data error.** Each user's recharge total matches their reward-tier flags exactly (all 5 users at the 100,000 tier actually spent 120,000+), consistent with the §12.4 tier counts.
4. **King's earns the most but from a few whales.** It is the single highest-revenue campaign yet depends on a handful of returning whales, so win-back budget should concentrate on Whale and High Value, and mean ARPPU should not be used in headline reporting.

12.6 Key Takeaways

1. **Dormant users were reactivated,** with a 64.92% return rate among eligible users.
2. **Campaign reach was high,** with 91.85% of eligible users exposed and 70.68% of exposed users returning.
3. **Reward systems performed well,** with consistently high login and recharge claim rates.
4. **Long-term retention was the weak point,** with the largest drop-off after reward collection.
5. **Value shaped outcomes,** as Whale and High Value users delivered far stronger retention and monetization.

12.7 Recommendations

1. **Prioritize high-value reactivation.** Whale and High Value users showed substantially stronger retention and revenue contribution; prioritize these segments when allocating win-back resources and testing differentiated incentives.

2. **Strengthen post-reactivation retention.** The largest drop-off came after reward collection; add follow-up engagement such as personalized push, daily missions, limited-time events, and VIP retention rewards.

3. **Develop segment-specific incentives.** Use low-barrier offers for Non-Payers, upgrade incentives for Medium Value, and exclusive benefits for High Value and Whale users.

4. **Test milestone-based retention mechanics.** High claim rates indicate strong engagement among users who reached each milestone; test whether progressive rewards can improve post-return activity and longer-term retention.

13. Fortune Recharge Analysis

13.1 Recharge Funnel Analysis

Questions

- How many users were exposed to the campaign?
- How many users visited the campaign page?
- How many users made a recharge?

- How many users earned Wealth Keys?
- How many users participated in the lottery?
- How many users reached the leaderboard?
- Where did the largest drop-off occur?

Results

Stage	Users	Conversion Rate	Drop-off Users
Exposure	1,800	-	-
Visit	700	38.89%	1,100
Recharge	183	26.14%	517
Earn Key	176	96.17%	7
Lottery Draw	161	91.48%	15
Leaderboard	20	12.42%	141

Findings

1. **Reach was broad but visit conversion was weak.** 1,800 exposures produced 700 visits (38.89%), leaving clear room to improve entry-page conversion.
2. **Recharge was the main funnel challenge.** 183 users recharged, a 26.14% visit-to-recharge rate.
3. **Post-payment engagement was very strong.** 96.17% of recharge users earned Wealth Keys, so reward qualification aligned well with behavior.
4. **Key holders used the lottery.** 91.48% of key earners drew in the lottery, showing strong interest in the reward loop.
5. **The largest drop-off was Lottery → Leaderboard.** Only 12.42% of lottery users reached the leaderboard, so the ranking layer appealed mainly to a smaller, highly engaged group.
6. **The reward system sustained engagement after recharge;** future work should focus on top-of-funnel traffic and recharge conversion.

13.2 Recharge Performance

Questions

- How many participants made a recharge during the campaign?
- What was the recharge conversion rate?
- How much revenue was generated?
- What was the average revenue per paying user (ARPPU)?
- How frequently did users recharge during the campaign?

Results

Metric	Result
Campaign Users	700
Recharge Users	183
Recharge Conversion Rate	26.14%
Total Revenue	164,294
ARPPU	897.78
Avg Recharge Days	4.01

Findings

1. **Recharging was repeated rather than one-off.** Paying users recharged on 4.01 days on average, indicating spending was spread across multiple days rather than a single transaction.
2. **Revenue came from a minority of participants.** 183 of 700 participants recharged (26.14%), generating 164,294 in revenue (ARPPU 897.78) — most revenue is carried by the paying minority, with depth-of-spend differences examined by value tier and by spend band.

13.3 Recharge Tier Analysis

Questions

- Which recharge tiers generated the most revenue?
- How concentrated was revenue among high spenders?
- Which user segments contributed the strongest monetization performance?

Results

Recharge Tier	Users	Revenue	Revenue Contribution	Avg Recharge Amount
No Recharge	517	0	0	0
Below 500	104	25,621	15.59%	246.36
500–999	41	30,578	18.61%	745.80
1,000–1,499	9	10,611	6.46%	1,179
1,500+	29	97,484	59.34%	3,361.52

Findings

1. **The Below-500 tier was the largest paying group** at 104 users, so the lowest spend band attracted the most payers by count.
2. **A few high spenders accounted for most revenue.** The 1,500+ tier was just 29 users but 59.34% of revenue.
3. **Top spenders far outspent the rest,** averaging 3,361.52 each — several times the other bands — which underscores the value of retaining them.

13.4 Key Distribution Analysis

Questions

- Which reward source generated the most Wealth Keys?
- How were Wealth Keys distributed across reward mechanisms?
- Which reward mechanism contributed most to the reward ecosystem?

Results

Key Source	Keys Earned	Contribution Rate
4-Day Recharge Reward	156	15.29%
Every 10K Recharge Reward	782	76.67%
Recharge Milestone Reward	82	8.04%

Findings

1. **Repeated recharge was the main key source.** The Every-10K reward generated 782 keys, 76.67% of all keys.
2. **Sustained recharge played a moderate role,** with the 4-Day reward at 15.29%.

3. **Milestone rewards were secondary,** at 8.04%.
4. **Recurring mechanisms beat milestone mechanisms** at driving reward accumulation.

13.5 Reward Mechanism Analysis

Questions

- How many users earned Wealth Keys?
- How many users participated in lottery draws?
- What percentage of key earners used their keys?
- How many users reached the leaderboard?
- How actively did users engage with the reward system?

Results

Metric	Result
Key Users	176
Lottery Users	161
Key → Lottery Rate	91.48%
Leaderboard Users	20
Lottery → Leaderboard Rate	12.42%
Total Lottery Draws	1,556
Avg Draws per Lottery User	9.66

Findings

1. **Key holders redeemed actively.** 161 of 176 key earners drew in the lottery (91.48%).
2. **Engagement was repeated, not one-off,** with 1,556 total draws and 9.66 draws per lottery user.
3. **The leaderboard was a narrow top layer.** Only 20 users (12.42% of lottery participants) reached it.
4. **The path from keys to lottery had little drop-off,** so redemption was intuitive and attractive.
5. **The lottery converted eligibility into participation,** while the leaderboard served as an advanced layer for core users.

13.6 User Value Analysis

Questions

- Which user segments generated the most revenue?
- Which user segments showed the strongest spending behavior?
- How did campaign performance vary by user value tier?
- Which segments should be prioritized in future campaigns?

Results

User Value Tier	Recharge Users	Revenue	Revenue Contribution	ARPPU	Avg Recharge Days
Medium Value	89	64,628	39.34%	726.16	3.96
Low Value	54	57,510	35.00%	1,065.00	4.00
High Value	33	31,631	19.25%	958.52	4.27
Non-Payer	1	7,476	4.55%	7,476.00	2.00
Whale	6	3,049	1.86%	508.17	3.67

Findings

1. **Medium Value was the primary monetization segment,** generating the most revenue (64,628, 39.34%).
2. **Low Value users upgraded their spend,** contributing 35.00% at an ARPPU of 1,065.00.
3. **High Value users recharged most repeatedly,** with the highest average recharge days (4.27).
4. **The Non-Payer figure is an outlier.** Only 1 user converted, so its 7,476 ARPPU is not a stable segment trend.
5. **Whales contributed little (1.86%),** so this campaign activated mid-tier users more than top spenders.(Whale recharge conversion is actually the highest, but with only 6 Whale users their absolute revenue contribution is small)
6. **Mid tiers drove this campaign;** stronger VIP-exclusive incentives may be needed to engage Whales. (Caveat: Whale and Non-Payer samples are very small, so read their ARPPU with care — recharge conversion, in §14, is the cleaner cross-tier measure.)

13.7 Lifecycle Stage Analysis

Questions

- Did participation and recharge-conversion rates differ across lifecycle stages?
- Did stage separate monetization as sharply as value tier does?

Results

lifecycle_stage	Stage Users	Participation Rate	Recharge Conversion
Active	1,667	24.5%	24.9%
At-Risk	591	20.8%	22.0%
Dormant	468	22.2%	**36.5%**
Churned	274	23.4%	25.0%

Findings

1. **Dormant users showed the highest recharge conversion (36.5%).** This is likely a data-timing artifact rather than a real pattern: the lifecycle_stage label is a snapshot, so some users tagged "dormant" had already returned and were spending again by the time the campaign ran. The group is also small, so this result should be read with caution.
2. **Stage separates monetization weakly.** By value tier, recharge conversion falls cleanly from 60% (Whale) to 0.7% (Non-Payer) — see the cross-campaign matrix in §15 — a much sharper split than stage gives here. For monetization, value tier or in-campaign spend bands are the better lens.

13.8 Leaderboard Analysis

Questions

- How different were leaderboard users from regular participants?
- Did leaderboard users spend more?
- Were leaderboard users more engaged with reward mechanics?
- What distinguished top-performing users?

Results

User Group	Users	Avg Recharge Amount	Avg Draws	Avg Wealth Coin
Leaderboard Users	20	3,829.90	35.95	817.35
Non-Leaderboard Users	680	128.96	1.23	20.42

Findings

1. **Leaderboard users spent far more,** averaging 3,829.90 versus 128.96 for non-leaderboard users.
2. **They engaged far more deeply,** averaging 35.95 lottery draws versus 1.23.
3. **They accumulated far more wealth coins,** 817.35 versus 20.42.
4. **Leaderboard users were the most valuable, most engaged group,** contributing disproportionately to revenue and activity.

13.9 Key Takeaways

1. **The campaign converted 183 payers** and generated over 164K in revenue.
2. **Revenue was top-heavy,** with the 1,500+ tier near 60% of the total.
3. **The reward loop engaged users,** with over 90% of key earners drawing in the lottery.
4. **Recharge-based rewards were the primary source of keys,** with the Every-10K Recharge reward accounting for 76.7% of earned keys.
5. **Leaderboard users stood out on both spending and engagement,** recharging ~29.7× more and making 30× more lottery draws than other users.

13.10 Recommendations

1. **Expand high-spender incentives.** The 1,500+ tier drove most revenue; add exclusive rewards for top spenders.

2. **Test stronger leaderboard incentives.** Leaderboard users showed substantially higher spending and engagement; test additional ranking tiers and exclusive rewards to determine whether stronger competitive mechanics can increase participation.

3. **Strengthen mid-tier progression.** Medium and Low Value users contributed a large share; add progression incentives to move them up.

4. **Optimize the reward loop.** The 91.5% key-to-lottery participation rate indicates strong downstream engagement; continue testing the connection between recharge, keys, lottery participation, and leaderboard progression.

14. SMS Reactivation (Always-on Win-back)

Beyond the four reward-mechanic campaigns, the platform runs an always-on SMS program to win back lapsed paying users (historical recharge above a spend threshold) who have gone inactive. Unlike the campaigns, it has no in-app mechanic or reward — it relies purely on message content to pull users back. Messages are sent in batches by lapse depth (3 / 7 / 14 / 30 days inactive).

Data note: this section uses real operational data, scaled to the project's sample size; all rates (delivery, reactivation, pay-through) are preserved.

14.1 Reactivation Funnel by Lapse Depth

Questions

- How does reactivation differ by how long a user has been lapsed?
- Does the share who pay after returning change with lapse depth?
- How cost-efficient is SMS reactivation, and where should spend go?

Results

Lapse Depth	Sent	Delivered	Delivery Rate	Reactivated (7d)	Reactivation Rate	Paid (7d)	Pay-through
3 days	2,852	2,029	71.1%	945	46.6%	240	25.4%
7 days	1,932	1,431	74.1%	443	31.0%	102	23.0%

Lapse Depth	Sent	Delivered	Delivery Rate	Reactivated (7d)	Reactivation Rate	Paid (7d)	Pay-through
14 days	585	449	76.8%	77	17.1%	17	22.1%
30 days	919	630	68.6%	51	8.1%	11	21.6%
Total	**6,288**	**4,539**	**72.2%**	**1,516**	**33.4%**	**370**	**24.4%**

Findings

1. **Reactivation falls sharply with lapse depth.** The reactivation rate drops from 46.6% at 3 days to 8.1% at 30 days — the longer a user has been gone, the harder SMS pulls them back.
2. **Pay-through is stable regardless of depth.** Around 22–25% of reactivated users pay within 7 days at every lapse depth, so the hard part is getting them back, not monetizing them once they return.
3. **Efficiency drops about 8x with depth.** Revenue per 1,000 messages falls from ~40,700 (3 days) to ~5,200 (30 days), so each message is far more productive on shallow lapses.
4. **SMS is cheap, but this is not causal ROI.** Revenue from reactivated payers (178,174) far exceeds the send cost (233), but with no holdout/control group these users might have returned anyway. Treat the figure as gross revenue coverage, not incremental lift.

Content angle

The messages relied on personalized, social-proximity hooks — a visitor viewed your profile, someone nearby is interested, an unread message is waiting — rather than discounts or rewards. For a social platform where the draw is connection rather than price, content-only reactivation fits the product and keeps cost near zero.

14.2 Recommendations

1. Concentrate spend on shallow lapses. Reactivation and revenue-per-send are highest at 3–7 days; treat 30-day sends as low-yield maintenance, not a growth lever.

2. Add a holdout group before scaling. A small untreated control would convert the current gross revenue figure into a true incremental lift, so reactivation can be attributed causally.

3. Use SMS and reward campaigns as a layered win-back. Always-on SMS handles broad, shallow, low-cost recovery; reward campaigns like King's Return handle deep, high-value dormant users. The two are complementary — wide-and-shallow versus narrow-and-deep.

15. Lifecycle Marketing Insights

The campaigns above were run at different times, targeted different user populations, and served different lifecycle objectives. They should therefore not be treated as controlled head-to-head comparisons. Instead, this section synthesizes recurring patterns observed across engagement, retention, win-back, and monetization use cases.

Insight 1 — Each lifecycle objective needs its own campaign design

Each campaign was built around a different lifecycle objective, and the observed results were consistent with that intent: the engagement mechanic drove participation, the check-in mechanic drove repeat visits, the reward campaign drove dormant-user return, and the recharge mechanic drove spend.

Lifecycle Objective	Lifecycle Focus	Campaign	What it primarily moved
Engagement	Active users	Puzzle Event	Participation & completion
Retention	At-risk / returning users	7-Day Check-in	Repeat daily visits
Win-back	Dormant users	King's Return	Reactivation & return
Monetization	High-value / paying users	Fortune Recharge	Recharge & revenue

The takeaway is structural: there is no single "engagement playbook" that serves every stage. A lifecycle program is a portfolio of objective-specific mechanics, not one campaign reused.

(The four campaigns ran at different times and were each designed for a different objective, so this is a design-to-outcome mapping, not a controlled head-to-head test.)

Insight 2 — User value was a consistently useful segmentation signal across campaigns

A recurring pattern across the campaigns was a clear performance gradient by historical user value. Higher-value tiers generally showed stronger engagement depth, conversion, reactivation, and retention, while lower-value and non-paying users showed weaker downstream performance. The replication matrix below shows one headline metric per campaign by value tier.

User Value Tier	Puzzle Completion	Check-in Day 7	Fortune Recharge Conversion	King's Return Rate	King's 30-Day Retention
Whale	100%	95.8%	60.0%	100%	91.7%
High Value	100%	73.4%	49.3%	91.3%	73.8%
Medium Value	79.8%	48.8%	46.1%	87.4%	31.4%
Low Value	44.1%	25.6%	19.2%	71.4%	2.7%
Non-Payer	4.5%	14.9%	0.7%	44.8%	0.0%

Notes: Puzzle/Check-in/Fortune use `user_profile` *tier; King's uses* `kings_user_profile` *tier (a separate population). For Fortune, recharge conversion is shown rather than ARPPU, because conversion lines up cleanly with tier while the §13.6 ARPPU is distorted by very small groups. Whale figures rest on small samples (6 in Puzzle, 10 in Fortune), so read them as directional.*

Historical user value was therefore a consistently useful segmentation signal across these campaigns. While the strength of the pattern varied by objective and population, value tier helped identify users more likely to engage deeply, convert, reactivate, or remain retained.

Insight 3 — Funnel bottlenecks differed by campaign, so optimization must be tailored

Each campaign lost users at a different point. There is no single drop-off to fix across the program.

Campaign	Main Bottleneck
Puzzle Event	Participation → Completion (only 37% finished)
7-Day Check-in	Gradual daily attrition (only 37.5% of unlocked users completed all 7 check-ins)
King's Return	Long-term retention (only 20% stayed to 30 days)
Fortune Recharge	Visit → Recharge (only 26% converted)

Because the binding constraint differs by objective, optimization should be tailored per campaign —improving completion mechanics for Puzzle, sustained engagement across the 7-day journey for Check-in, post-return retention for King's, and recharge conversion for Fortune — rather than applying one engagement fix everywhere.

Insight 4 — Revenue was highly concentrated among top spenders

Revenue concentration appeared clearly in the monetization-focused analyses: Fortune's top spend band (29 users) generated 59% of campaign revenue; King's Whale and High-Value users generated 94% of campaign revenue; and Fortune's 20 leaderboard users recharged about 30× more on average than other users. These patterns suggest that protecting and re-engaging high-value users can have disproportionate revenue impact.

Insight 5 — Win-back works in layers

Reactivation is not one campaign but a layered system. Always-on SMS recovers lapsed users broadly and cheaply, but its pull decays fast with lapse depth (47% at 3 days to 8% at 30 days); the King's Return reward campaign recovers a narrow, high-value dormant pool deeply (65% reactivation). The two appear to serve complementary roles in the win-back strategy — SMS for broad, early, low-cost recovery and incentive-based campaigns for deeper or high-value dormant users — rather than functioning as direct substitutes.

Methodology note: Campaigns ran at different times and targeted different populations and objectives. Cross-campaign observations are directional business patterns rather than controlled performance comparisons.